



MID-TERM SHELTER SOLUTIONS FOR FORCIBLY DISPLACED CITIZENS OF MYANMAR



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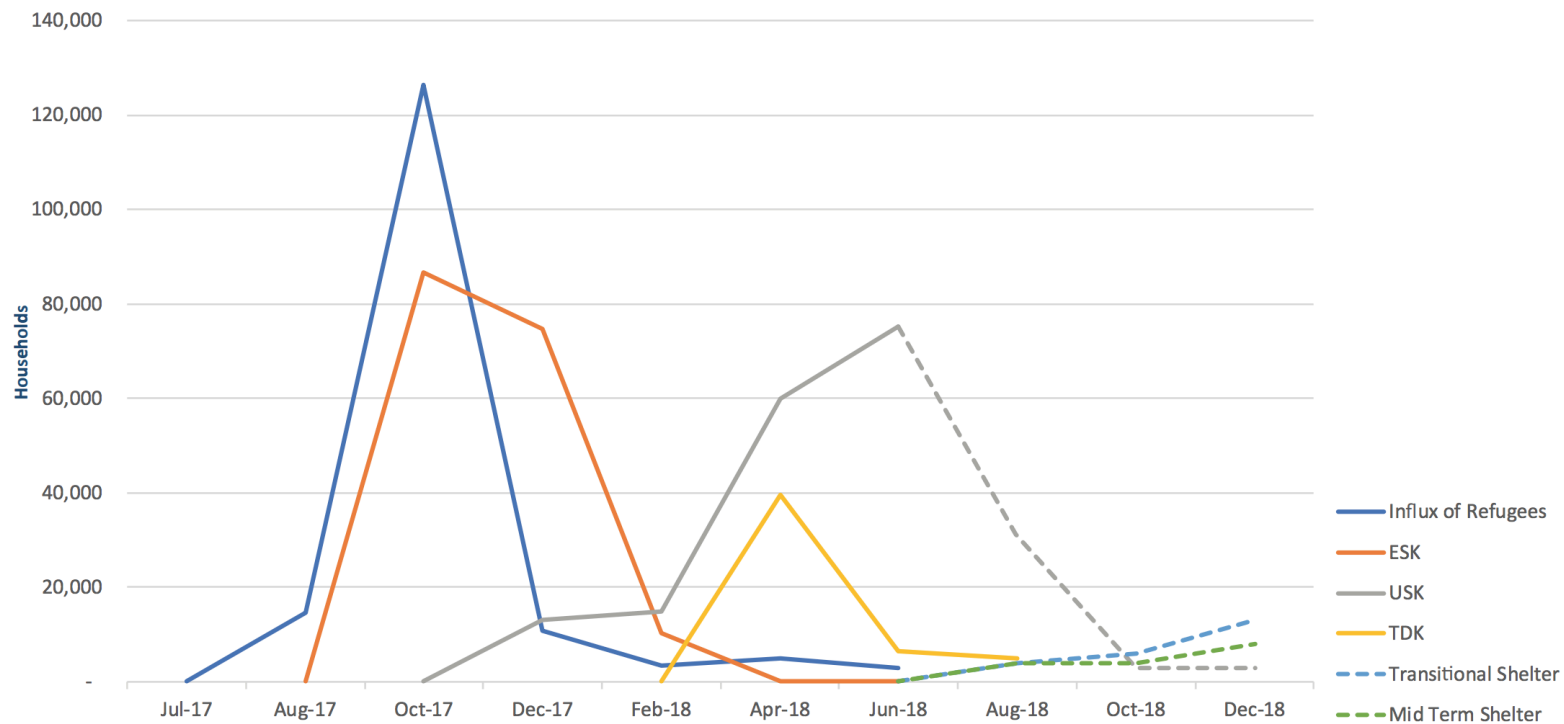
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CONTEXT



Shelter Options - Time-lapse



Mid-term Shelter TWiG



Save the Children

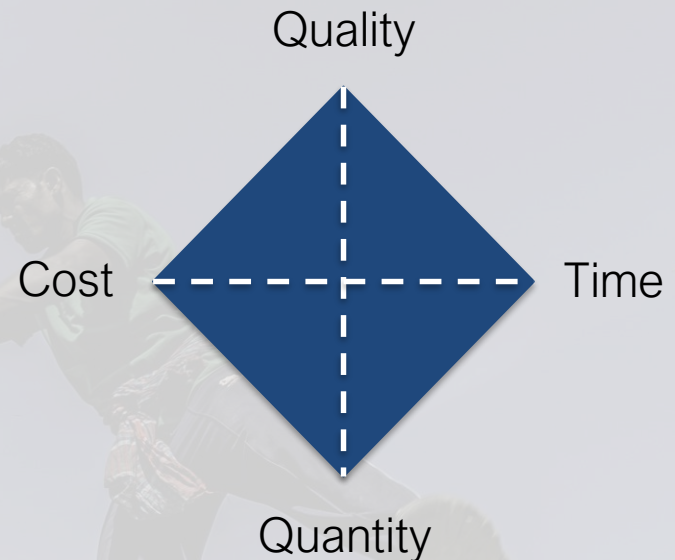
DESIGN PRINCIPLES AND DRR FEATURES



Shelter Design Considerations

Designs should:

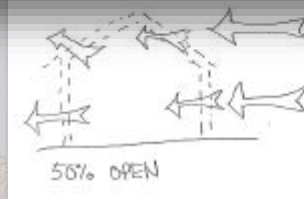
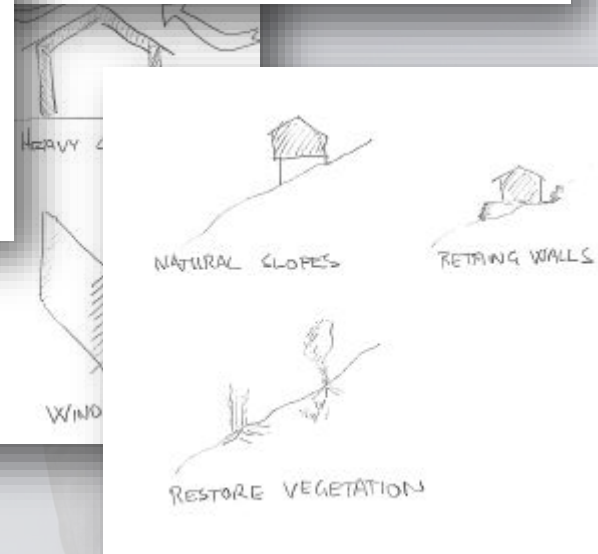
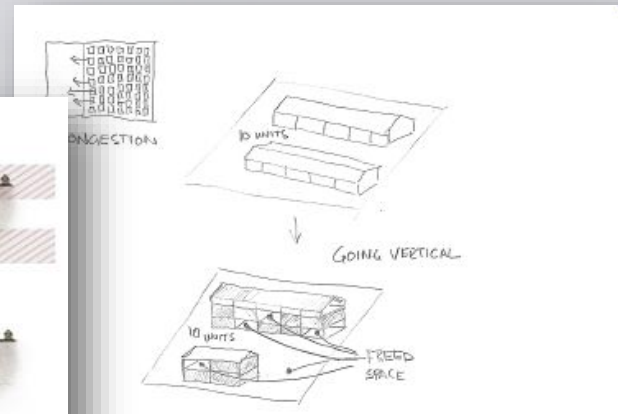
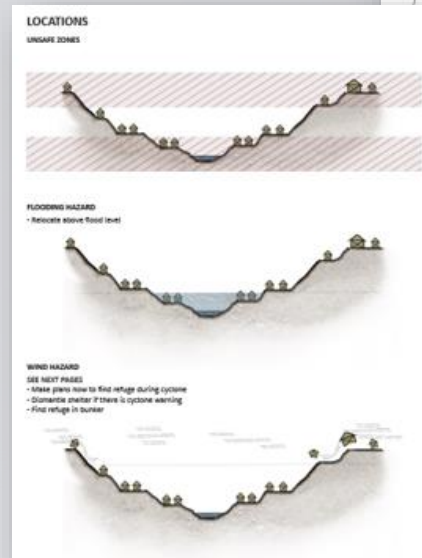
- Provide better durability in accordance to Phase 3
- Benefit both the local economy and host communities
- Ensure household and community participation
- Aspire to meet relevant standards – Govt / SPHERE
- Be cost effective
- Be flexible
- Be appropriate to climate and weather
- Make use of existing capacity



Design Constraints and Considerations

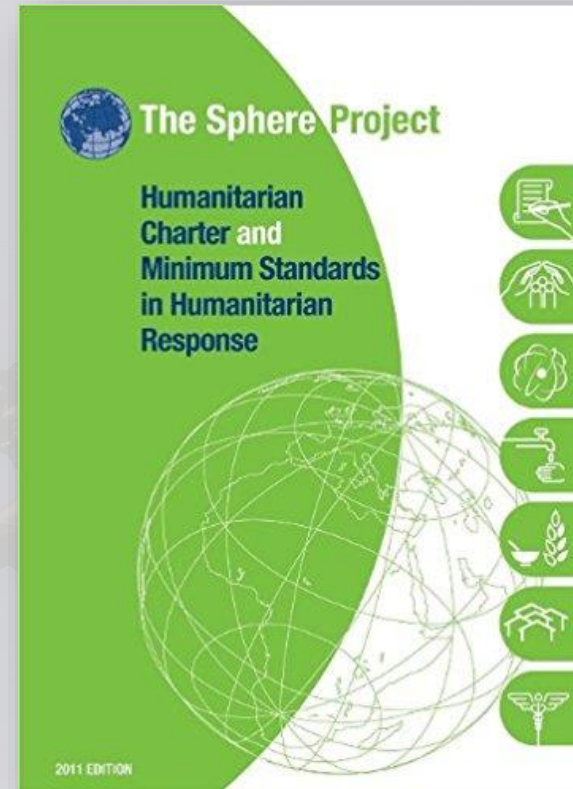
- Congestion
- Flood
- Landslide
- Cyclone
- Fire
- Materials

Most of the issues above relate to the need for **space**.



Standards - SPHERE

- Benefit to both Host and Rohingya Communities
- Maximize **Participation**
- Covered living space - **3.5m² / Person**
- Safety and Security
- **Dignity** – culturally appropriate
- Minimize environmental impact



Protection for the environment

- Designs should **not further impact the environment**
- Shelters should be constructed mainly from **organic and natural materials** that will decompose when the site is vacated and reinstated
- Non organic materials, such as concrete posts, should be **removable and re-usable for hosting communities**



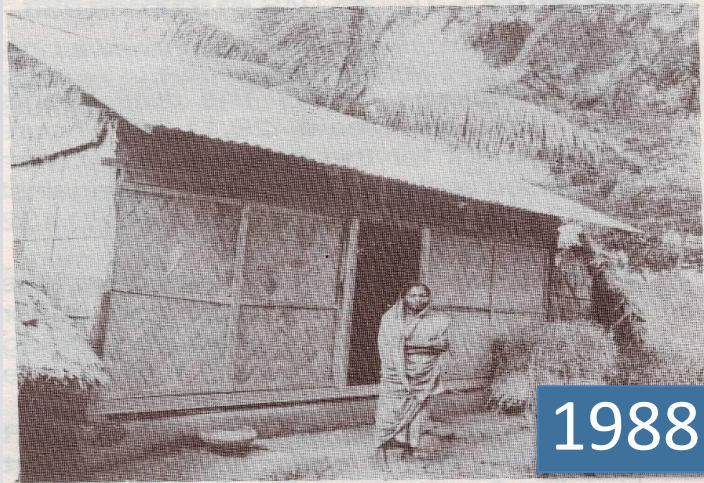
Economic and Social Benefits

Approaches and designs should :

- **Benefit the local economy** and encourage good relations between Host and Refugee communities.
- Capitalize on the **high level of local skills and labour availability** in the Rohingya and Host community
- Make use of **local and regionally available materials** and manufacturing capacity.
- Support **local business and markets**



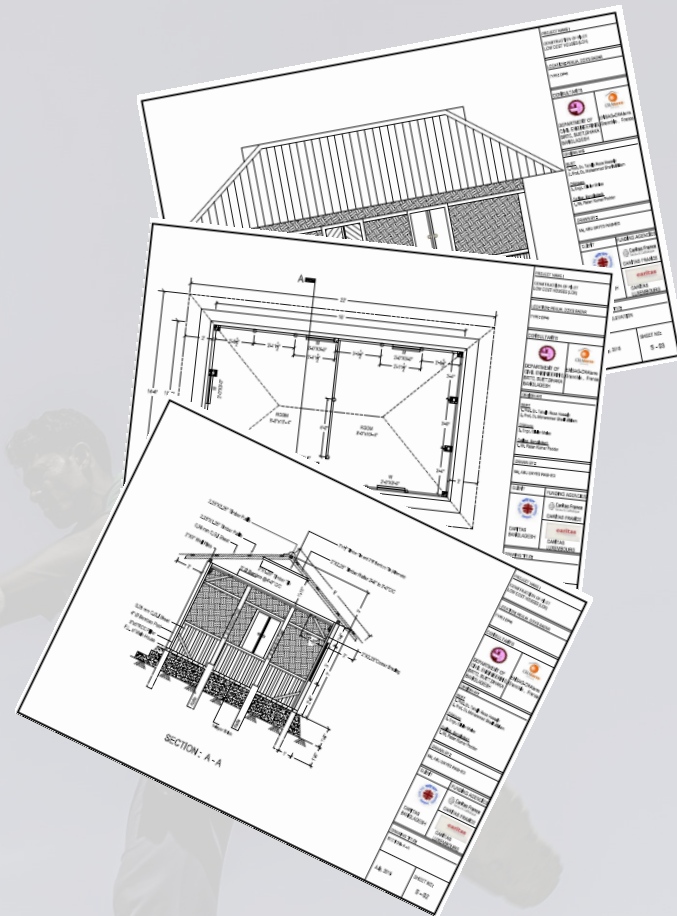
Existing knowledge for low-cost housing



Tried and Tested

Existing engineered designs available:

- Already implemented in the area
- Designed to 120kph wind speed
- Simple construction techniques
- Using local materials
- Designs to follow principles for cyclone coastal region – hipped roof





Shelter built in 2015 at Pekua, resisted cyclone Roanu and Mora



Construction

- Pre-caste concrete posts - Strong and durable
- Can be used with bamboo or timber frame
- Different wall covering materials can be used
- Can be manufactured locally – providing local employment
- Concrete post can easily be removed and re-used when the shelters are decommissioned
- Allows for further upgrading
- Allows for modification to the design if required



Designs approved by the RRRC (local government)

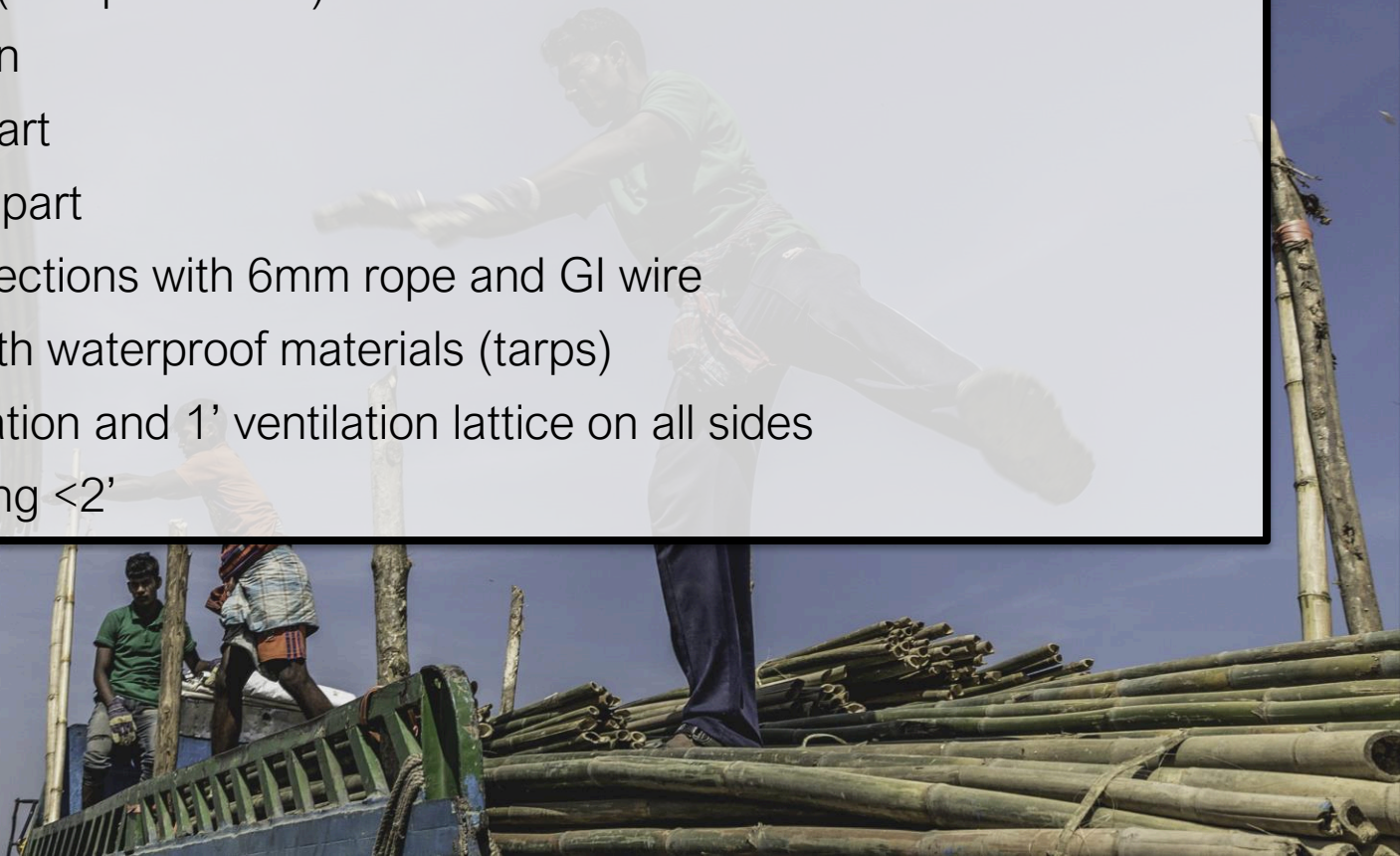
- 5 designs with 4 different dimensions
- Same construction method – RCC posts + bamboo posts, bamboo wall frame and roof frame
- Different wall cover material – bamboo fence, CGI sheet

Option	Length	Width	Height	Floor Area (ft ²)
1	18'	10'6"	7'	189
2	16'	12'	7'	192
3	14'	14'	7'	196
4	18'	10'6"	7'	189
5	15'	10'	7'	150

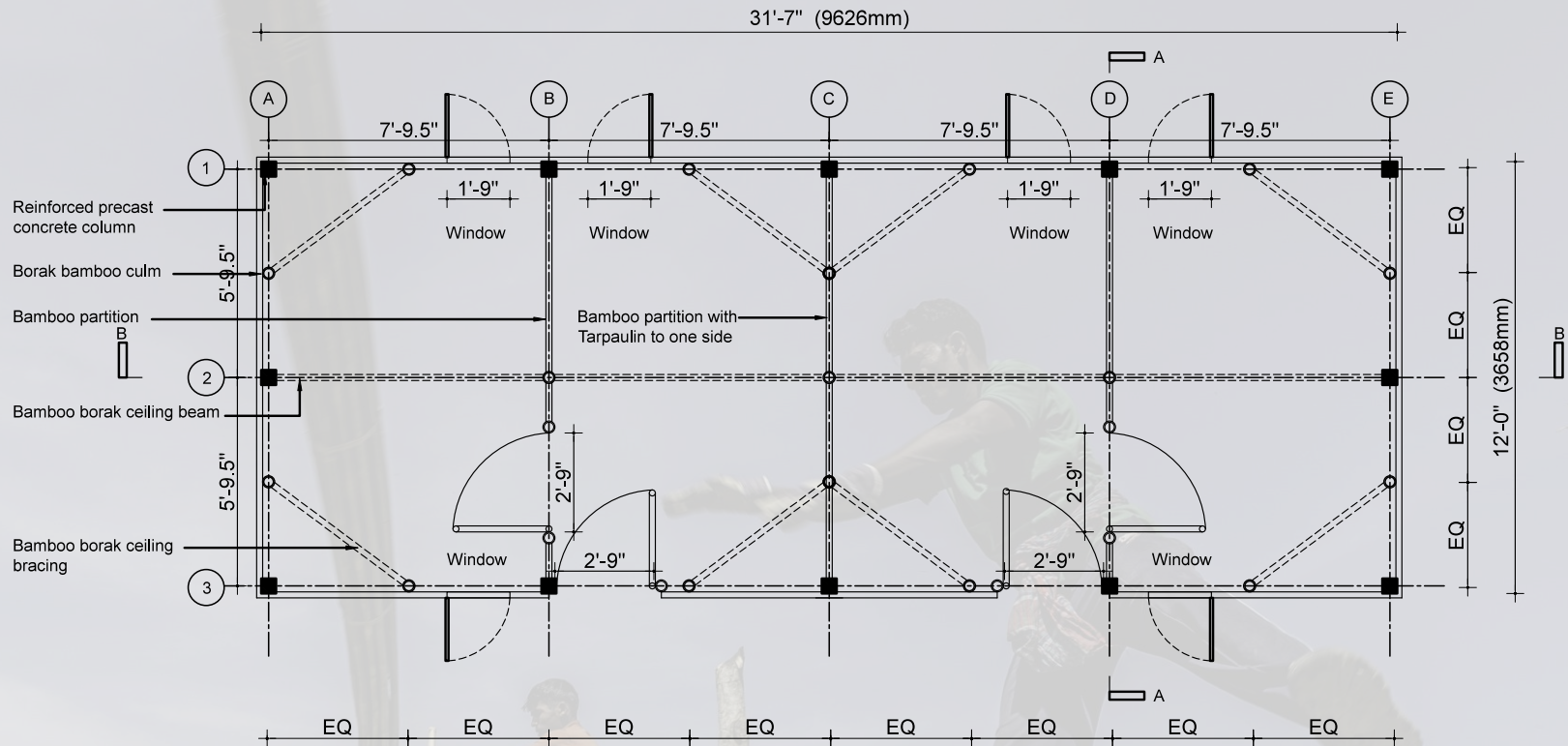


Main design principles

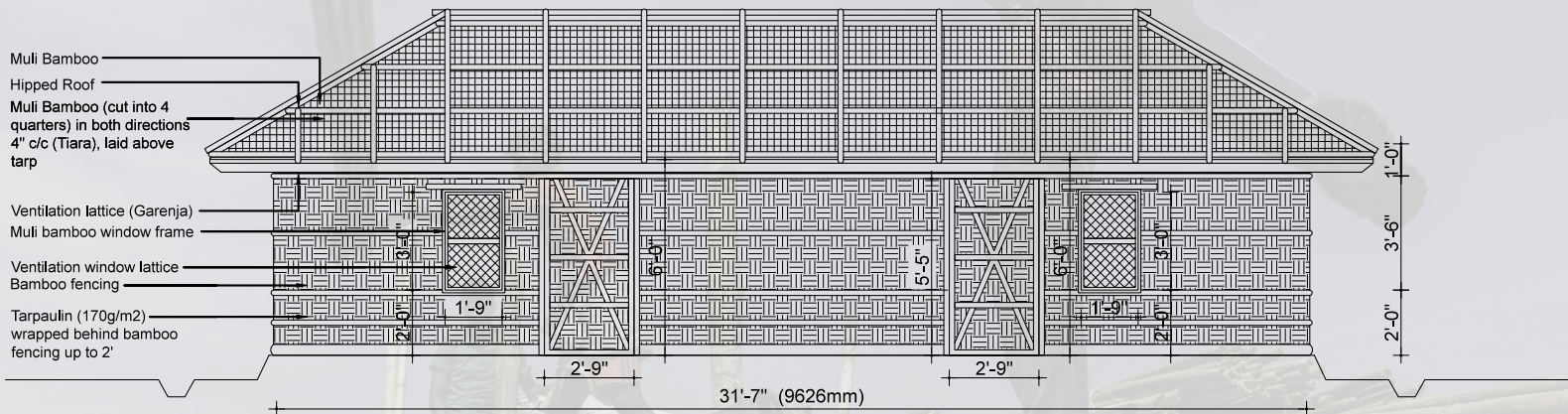
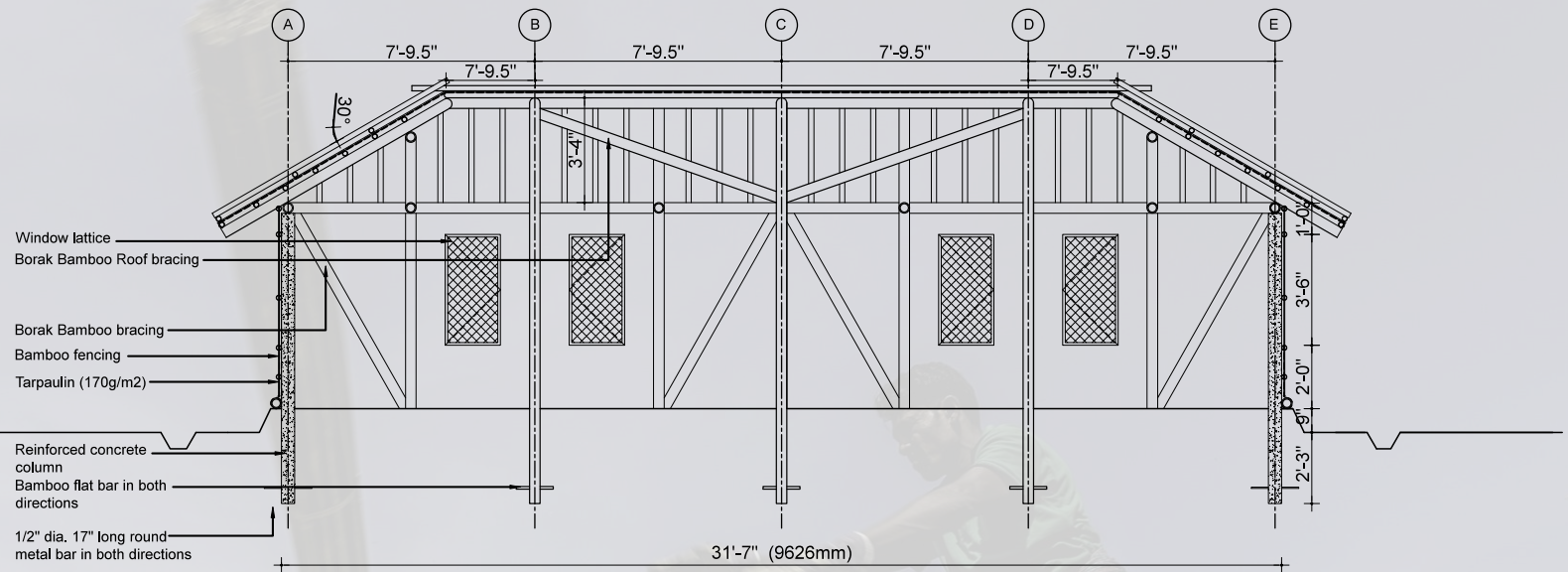
- Raised plinth
- Cross anchorage (2'-6" deep) to improve resistance to uplift
- Bracing at all levels (vertical, ceiling, truss and roof)
- Hipped roof (roof pitch $>27^\circ$)
- Roof tie down
- Posts $<5'$ apart
- Rafters $<1'$ apart
- Strong connections with 6mm rope and GI wire
- Bottom 2' with waterproof materials (tarps)
- Cross ventilation and 1' ventilation lattice on all sides
- Roof overhang $<2'$



Designs approved by the RRRC (local government)



Designs approved by the RRRC (local government)



PILOT PHASE



Model Houses Constructed

- Different wall bracing systems
- Different roof bracing systems
- Different wall cover materials
- Different roof cover materials and systems
- Different roof overhang

Option 2



Option 1

Option 3

Option 4

Option 5



Model Houses Constructed

Option 2 with RCC posts, double house



- Option 1 as per drawings with some improvements learnt from the first 5 models
- Option 2 double house
- Different wall cover materials
- Different doors and windows materials

Option 1 with RCC post



Model Houses Constructed

- 2 model houses built with footings in collaboration with UNHCR.
- Footing design by UNHCR
- Bamboo posts above footings, no RCC posts
- Different floor height (6'6" and 6' instead of 7')
- Different wall cover materials

Option 3 with pad footings



Option 5 with strip footing



Wall Bracing Improvement

Original design: weak bracing

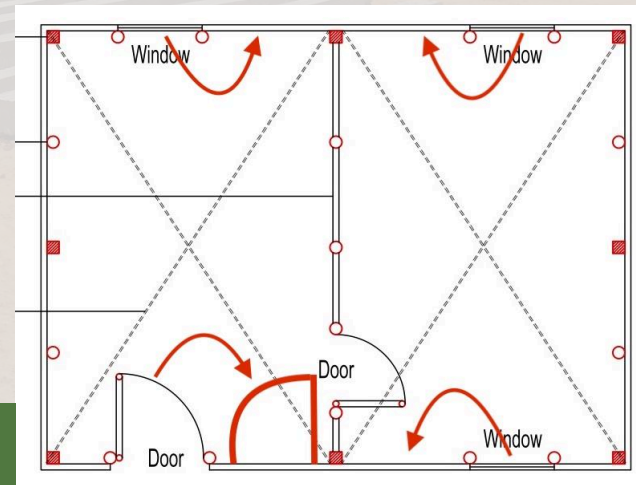


Strong bracing at both ends



- Steep and weak bracing in the original design
- Relocate the door and windows
- Make strong braced frame at both ends of the house

Relocate door and windows



Roof Overhang

Community Structures Design and Strengthening Guide

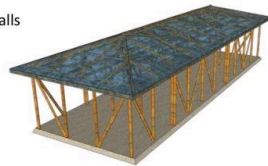
SECTION 1: DESIGNING NEW BUILDINGS

ESSENTIAL REQUIREMENTS:

1. Not in flood zone
2. Not in landslide zone
3. Drainage around building
4. Good access
5. Plinth
6. No bamboo embedded in ground
7. Columns anchored into ground
8. Beams tied down to columns (with dowel or through column)
9. Borak bracing in all four walls
10. Maximum 20' between braced walls
11. Perimeter Column spacing < 5'
12. Internal Column spacing < 10'
13. Rafter spacing < 5'
14. Rafter span < 8'
15. Purlin spacing < 1'
16. Rope bracing in roof
17. Roof is tied down

PREFERABLE REQUIREMENTS:

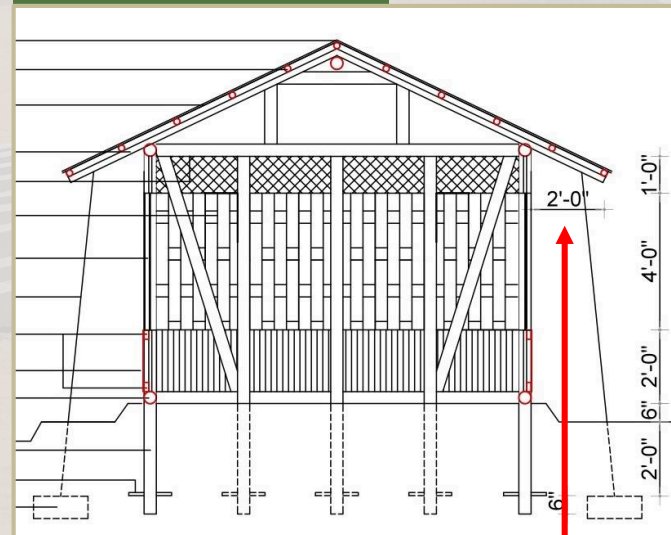
1. Hipped roof
2. Small roof overhang (cut back if greater than 1.5')
3. Sheltered from wind
4. Good access during rains
5. Male and Female WASH facilities



- ✓ Hipped roof reduces wind load
- ✓ Roof tied down to columns
- ✓ Columns anchored into ground
- ✓ Perimeter columns at 3-5' spacing
- ✓ Internal columns at 6' - 10' spacing reduce span

Small roof overhang (cut back if greater than 1.5')

Design drawing



Roof overhang = 2' for all options

- Anomaly between guideline and design drawings
- Size of overhang – compromise between wind uplift forces with shading and rain protection

Roof Overhang



Option 1, 4, 5:

Overhang of 1'6" is sufficient for rain protection of bamboo ventilation Garanja



Option 2:

Overhang of 1'9" is sufficient for rain protection of bamboo ventilation Garanja



Option 3:

Overhang of 2' is sufficient for rain protection of bamboo ventilation Garanja at the front and back of the shelter

Layout Dimensions: Summary

Option	Floor Area (ft ²)	Floor Area (m ²)	Floor Area m ² / Person	Remarks
1	189	17.6	3.52 (5 people)	• Most expensive option due to 2 extra RCC posts • Least strong frame
2	192	17.8	3.56 (5 people)	• Most economic option • Strongest frame
3	196	18.2	3.64 (5 people)	• “Deepest” building, therefore longest roof span and partition wall
4	189	17.6	3.52 (5 people)	• Same as Option 1
5	150	13.9	3.48 (4 people)	• Smaller building, not meaningful to compare with other options

Cost of Model Houses

Option	Cost (BDT)	Remarks
1	78,800	It is not too meaningful to compare these cost because: - each option may vary in wall cover details - each option may vary in roof cover details - each option may have a different roof overhang - all options have no RCC posts
2	78,600	
3	73,500	
4	74,500	
5	75,200	
1 - RCC	101,500	Original budget = 60,000
2 - RCC Double	146,900	= 73,500 per house
3 - Pad FTG		To be confirmed
5 - Strip FTG		To be confirmed

ROLL-OUT AND NEXT STEPS



Camp 20 Extension

- New expansion area - IOM Area of Reference
- Construction of **305 Mid-term shelter units** to relocate families at high risk of natural hazard



Camp 20 Extension

- Relatively remote area – logistic challenges – RCC posts casted on site
- Pre-assemble building components



Camp 20 Extension

- Both host and refugee communities involved in the construction
- Single and double chamber shelters



Camp 20 Extension



Camp 20 Extension



Camp 20 Extension



Camp 20 Extension



Camp 16 - CARE



Camp 16 - CARE



Camp 16 - CARE



Next steps

- Identify procedures to implement **Mid-Term shelters** in existing camps, working closely with Site Management
- Identify vulnerability criteria and participatory processes for targeting and prioritization
- Explore options in case existing areas are too dense and do not reach minimum standards
- Prepare training modules to assist families in the implementation
- Link shelter construction with livelihood opportunities (in particular inclusion of women)
- Identify procedures to upgrade existing shelters to Mid-term

Thank you!

